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Degree Thesis

Initial Investigation Study

Changes of metacarpal region temperature in horses: comparison of the use of germanium and carbon bandages with that of ice.

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Summary

Introduction and goal: Incrediwear bandages are impregnated with germanium and carbon. Based on their composition, if wet, they are assumed to have a cryotherapy effect. The cryotherapy method most commonly used in sporting horses is the application of ice to the distal part of the limbs, inside protectors, as a way to control possible injuries induced by the training of the flexor tendons. This study comparatively assesses the changes in skin temperature induced by the two procedures (Incrediwear bandages vs ice) in the metacarpal region. Methods and materials: In a total of 10 horses, an Incrediwear bandage was applied to one forelimb and an ice protector to the other. They were kept on for 20 minutes. The maximum temperature (Tmax) was measured using thermography before the application of the two systems, immediately after removing them, and at 5, 10, 15, and 30 min. afterwards. Results: Both the systems induced a significant reduction of Tmax compared to the values before application, with Tmax remaining reduced during the first 30 min. after removal. The Incrediwear bandages induced a lower Tmax during the first 10 and 15 min. after removal, compared to the ice protectors. Conclusions: The application of bandages impregnated with germanium and carbon in the metacarpal region of the horse results in a more intense reduction of Tmax than the application of ice protectors.

Keywords: Cryotherapy, Horse, Exercise, Tendinopathy.

1. Introduction

Exercise requires the involvement of various systems of the body, creating stressful stimulation for the athlete. The changes caused by exercise include the reduction in the muscular energy reserve, rise in metabolic waste, inflammatory reactions in soft tissues, and microscopic lesions in tissues, stimulating the cycle of their repair and rebuilding (Boffi, 2006).

Riders often use various types of bandages, such as working bandages, post-exercise bandages, and rest bandages, in order to control movement, reduce the inflammation caused by exercise, and minimise the risk of injuries. Some authors believe that their use during exercise promotes an increase in the internal temperature in the flexor tendons, primarily in the superficial digital flexor tendon (SDFT), which could lead to the degeneration of tenocytes. (Patterson-Kane et al., 2009; O'Brien et al., 2021). The hyperthermia during exercise has partially been attributed to the low vascularisation of these anatomical structures, which limits the dissipation of the accrued heat. This has been viewed as one of the pathophysiological mechanisms involved in SDFT tendinopathies in horses (Birch et al., 1997; Burrows et al., 2008).

Meanwhile, contrast therapy with heat and cold is commonly used for post-exercise recuperation in human athletes. The application of heat is used primarily to increase the pliability and flexibility of soft tissues while reducing pain and muscular hypertonicity. The purpose of cryotherapy, i.e., the application of cold, is to reduce inflammation and pain (Nadler et al., 2004).

Despite this, the effect of cryotherapy has not been demonstrated in the application of bandages in horses and the occurrence of musculoskeletal injuries.

2. Prior studies in the field

2.1. Cryotherapy in sports horses

Cryotherapy, or the therapeutic use of cold, is a common practice in human sports medicine, and its purpose is to stimulate vasoconstriction, reduce metabolism in the cells (those that can maintain their viability in the conditions of a lesion, thus avoiding cell death), minimise the formation of oedemas, and lower pain after exercise (Apolo Arenas et al., 2015). It is believed that in humans whole-body cryotherapy (WBC) can produce analgesia thanks to the reduction of nerve conduction and suppression of pain signals to the dorsal horn of the spinal cord. Furthermore, it is associated with the stimulation of cold reflexes, such as vasoconstriction, deceleration or even blocking of nerve conduction, which

ultimately contributes to pain relief (Bogard et al., 2020). In fact, regional hypothermia has shown promise as an analgesic in horses when used for managing painful conditions of the distal extremity, such as laminitis (Raingera and Wardiusb, 2020). Recent studies of humans have demonstrated that exposing the body to extremely low and high temperatures leads to the production of free radicals (highly unstable molecules causing cellular damage) (Bogard et al., 2019). For this reason, it is not recommended to apply too much cold, because an excess of extreme vasoconstriction will impede the elimination of all the substances created by exercise that have negative effects on tissue.

Thus, it is recommended to apply cold after exercise, especially if there is no acute damage. In order for the procedure to be correct, first, one must take account of the temperature that the tissue must be exposed to for the beneficial effects of cryotherapy to be achieved. As regards what temperature needs to be achieved, there is evidence that lowering the surface temperature of the distal extremity to less than 7 °C reduces the response to mechanical stimulation. Once the surface temperature in the extremity falls below 7 °C, every additional degree of temperature reduction leads to an increase in the force necessary to provoke a response (Raingera, 2020).

Another aspect to be considered is the procedure of application itself, which involves the time that the tissues are to be exposed and the manner of application, which can be general, covering the entire body (for human athletes), or local, involving the extremities, the thoracic limb up to the carpus and the pelvic limb up to the tarsus (for horses). In terms of time, the requirement is a minimum of 10 minutes of application and a maximum of 20–25 minutes, to avoid the vasodilation rebound effect (Bogard et al., 2019). In humans, the form of cryotherapy increasingly common among elite athletes is WBC, with cryogenic chambers specifically designed to expose the athlete to temperatures below -100 °C, which is thus an example of general cryotherapy, as opposed to local in horses. The phenomena associated with the beneficial properties of cryotherapy, such as the reduction of pain, are observed in humans once the muscular temperature falls to -27 °C. Currently, given the absence of scientific guidelines for horses, experimental WBC systems can only be based on the same concept as that used on humans. The entire body of the horse suffers a thermal shock at -140 °C, caused by exposure to gaseous nitrogen in a semi-enclosed space for a limited period

(usually 3 minutes). In a study by Bogard et al. (2020), which researched the use of general cryotherapy in horses, it was observed through thermal imaging that 3 minutes of exposure to a temperature of -140 °C was not sufficient to create a sufficient thermal gradient in the skin of the horse. As a result, this treatment protocol failed to cause the benefits of the cold

to develop. It is necessary to conduct more studies with the goal of developing a protocol that is appropriate to specifically treat horses.

For horses, there are various procedures for reducing the temperature locally. In general, they cover the distal part of the extremities, primarily because injuries of the tendons and ligaments have a higher incidence than muscular damage, and because of evolutionary modifications in horses, which led to a reduction in muscle mass in the distal part of the horse's extremities. Cold water showers could be a good option as local cryotherapy, even if the water does not reach temperatures that are particularly low. Also, one can apply ice locally, in an area with an acute or chronic injury: keeping in mind, however, that the best results of cryotherapy are gained whenever it is used between 24 and 72 hours after the damage occurs. Furthermore, one must take care to avoid damaging the skin, for which reason it is recommended not to apply the ice directly to the skin. Another option is to perform the treatment in the form of massage with ice.

In a study by Wilson et al. (2018), they conducted surveys to determine the main therapies used in the sports horses that participate in international competition. They found that the application of ice was performed in 95.2% of cases, compressive bandages were used in 89.5% of cases, and hydrotherapy with cold water, in 82.9%.

2.2. Bandages and sportswear with germanium and carbon

In recent years, bandages manufactured out of a special fabric have begun seeing use in humans. Specifically, this fabric consists of a fibre impregnated with germanium and carbon, which activate with exposure to body heat, producing negative ions that cause molecular vibrations and leading to a mechanical increase in blood flow and lymphatic drainage in the area. The rise in blood flow and its speed, and in lymphatic drainage, help bring more oxygen and nutrients to the target area and eliminate the waste products (Hamblin et al., 2017).

The difference between semiconductors like germanium and metals is that whenever the temperature of a semiconductor rises, its resistance falls because of a larger quantity

of free electrons, enabling better conductivity. It has been thus suggested that when electrons are released in a particular direction, due to elevated temperature, germanium can have transdermal properties (Lee et al., 2018; Marino et al., 2019).

2.2.1 Germanium

On a systemic level, as germanium is introduced into the body, it attaches to haemoglobin, stimulating cell aerobic metabolism (Kobayashi y Ogra, 2009). At the same time, it helps in removing hydrogen ions and other harmful radicals. The elimination of free radicals improves the antioxidant capacity of the body, leading to a reduction in lipid peroxidation and protecting cell membranes (Liu et al., 1997). It increases the expression of the IL-3 cell stimulator factor in the haematopoietic pluripotent stem cells, participating in the regulation of their spread and differentiation, which has a beneficial effect on haematopoiesis (Schwartz et al., 1983). Its antibacterial effect has been demonstrated with many infectious agents, such as *Bacillus subtilis*, *Staphylococcus aureus* or *Escherichia coli* (Dyke et al., 1989). It improves non-specific immune, cellular, and humoral activity, and promotes the synthesis of cytokines (Suzuki et al., 1985; Li et al., 1991).

2.2.2 Far infrared radiation

Germanium emits elevated levels of FIR, far infrared radiation, the biological effects of which have been known for a long time, especially its interactions with various biological structures such as cell membranes and fluids, primarily water, DNA, and proteins (Ji-yeon et al., 2017). The biophysical mechanisms underlying the interaction of FIR with cells are based on changes in action potentials and in mitochondrial metabolism (Sheppard et al., 2008). FIR is absorbed by water molecules and other charged parts of the cell, such as proteins, resulting in rotation and vibration. These effects depend on temperature, which significantly affects the absorption of such radiation by biological systems (Ji-yeon et al., 2017). Thus, it has been suggested that the application of FIR results in an increase in metabolism and blood circulation (Vatansaver and Hamblin, 2012; Lin et al., 2013), and in tissue regeneration (Segovia et al., 2011), as it causes elevated production of nitric oxide dependent on calcium and the release of calmodulin in different cellular lines (Leung et al., 2009).

Based on these findings, it has been assumed that FIR has beneficial effects thanks to its antioxidant, analgesic, and anti-inflammatory activity (Vatansever and Hamblin, 2012; Nunes et al., 2019).

Given these properties of germanium and resulting FIR, different types of functional gear and athletic clothing impregnated with particles of this semiconductor have been developed (Ji-yeon et al., 2017). According to Chu et al. (2015), clothes impregnated with germanium control the thermal stress caused by exercise through improvements in blood circulation. Later, Ji-yeon et al. (2017) examined the effect of so-called 'functional clothing' that contained germanium in terms of anaerobic capacity, fatigue, and oxygen transport in 10

individuals, conducting two exercise tests in the same athletes at an interval of two days. The number of blood cells, the concentration of haemoglobin, and the red cell distribution width rose significantly during exercise in clothing impregnated with germanium and did not when the athletes wore their normal clothing. These authors suggested that the use of clothes treated with germanium could improve the capacity for transporting blood oxygen during short-duration, high-intensity exercise (Ji-yeon et al., 2017).

2.3. Studies on the use of bandages that emit or reflect far infrared radiation in horses

Based on the bandages intended for humans, the company Incrediwear® designed bandages for horses. According to the company, the bandages help preventing injuries, speed up recovery, and improve performance (<https://incrediwearequine.com>). Once soaked in water, they can be used for exercise, for recovery after injuries, and quick cryotherapy.

The manufacturer emphasises benefits such as lower swelling and inflammation, reduction of pain and muscular fatigue, faster recovery, better blood circulation and lymphatic drainage.

The studies researching the use of the bandages in horses (both during and after exercise) are few, and the opinions on their use vary. It has been stated that bandages for tendons can improve anatomical support, reducing the maximum extension of the metacarpophalangeal/metatarsophalangeal joints and protecting the extremities from injuries. Additionally, it has been believed that bandages could be helpful in reducing uncoordinated movements caused by fatigue and disruptions in proprioception (Koblunk et al., 1988; Smith et al., 2002; Kicker et al., 2004; Westermann et al., 2014).

Westermann et al. (2014) measured the skin temperature when taking thermal images before and after exercise, under various conditions: with the front extremities without bandages, and

later, a bandage impregnated with ceramic particles applied to the left metacarpal region and a tendon protector applied to the right metacarpal region. The exercise performed by the horses consisted of 20 minutes of exercise with the horse tied (10 minutes' walk and 10 minutes' trot, with each of the hands). The authors noted that the skin temperature did not change significantly during the exercise if the limb was not bandaged (the average temperature before exercise was 14.1 °C, and after, 14.4 °C). The temperature rose from

15.3 °C at rest to 24.8 °C with the bandage, and from 15.3 °C to 20.6 °C with the protector. Westermann et al. (2014) suggested that the use of bandages could accelerate the warming-up of the connective tissue during exercise.

The consequences of the increase in temperature caused by the bandages applied in the metacarpal region are not clear, with different opinions. A previous study (Westermann et al., 2013) found that the flow of air was effective at cooling down the skin of the distal part of the limbs. According to some authors, minimising such losses could have a beneficial effect, especially in exercise that takes place at a low ambient temperature, which can cause vasoconstriction (Westermann et al., 2013). In fact, they studied the influence of ambient temperature on skin temperature, observing that with high ambient temperature, the temperature of the distal part of the limbs was also high, with the horse at rest (Westermann et al., 2013). Some researchers suggest that the increase in skin temperature, which could also affect deeper structures, could be beneficial for the tendons of the metacarpal region. A moderate increase in blood flow could help reduce the risk of injury (McCutcheon et al., 1999). Meanwhile, those disputing this theory state that the accumulation of heat in the interior of the tendon could be a risk factor in the development of tendinopathies (Wilson and Goodship, 1994; Birch et al., 1997; Farris et al., 2011). It has been suggested that there is little vascularisation in the core of the tendon, which means that the removal of accumulated heat could be compromised. The resulting hypoxia and deterioration of cell metabolism could have adverse effects on the tenocytes.

3. Goals and hypotheses

On its website, the company that manufactures and distributes bandages impregnated with germanium and carbon for use on horses (<https://incrediwearequine.com>) shows a few studies, but none of them is a standardised investigation, and none has been published in a peer-reviewed journal.

Thus, the **principal goal** of this study is to analyse the difference in temperatures arising from the use of bandages impregnated with germanium and carbon in sports horses, as opposed to the use of conventional ice protectors.

The area of the treatment is the metacarpal region. It was selected due to the elevated incidence of SDFT tendinopathies. This tendon accumulates energy, which makes it very susceptible to damage during activities such as galloping and jumping, for which reason SDFT tendinopathies are a common cause of lameness in sports horses, primarily in thoroughbred racing and show jumping horses (Perkins et al., 2005; Avella et al., 2009; Thorpe et al., 2010).

Our **initial hypotheses are**: first, the use of bandages impregnated with germanium and carbon, soaked in water, with the horse at rest, results in a reduction of the surface temperature in the metacarpal region of both tendons; second, the reduction in the temperature is more intense with the use of Incrediwear bandages compared to the use of an ice protector, which is a system used conventionally by coaches and riders.

4. Methods and materials

This study was carried out under the directives of the University of Cordoba Committee of Bioethics and Biosecurity in Animal Experiments. Given the routine and non-invasive nature of the proceedings in treating sports horses, no approval by the ethics committee was required. In all cases, the owners of the horses accepted the inclusion of their animals in the study.

4.1. Horses

10 adult horses were studied, 7 male and 3 female, with ages ranging from 7 to 11 years old. In terms of the breeds, there were 4 Spanish Sport Horses, 1 Belgian Sport Horse, and 5 Dutch Warmbloods. All of the horses competed in show jumping: 3 at 1.50 m, 3 at 1.40 m, and 4 at 1.30 m.

The number n of the horses was calculated using the GPower 3.1 software, assuming as the final point a T_{max} reduction of 5 °C, starting with the baseline values obtained before the application of the systems. The number n of the horses was set to 10, for a power of 80% and an alpha value of 0.05.

No special criteria were followed in the selection of the animal, except for making sure that they did not have a clinical history (old or recent), tendinopathies, or other types of injuries in the front limbs, with a special emphasis on the metacarpal region. Figure 1 shows one of the horses that participated in the study.



Figure 1. One of the studied horses during a show jumping competition.

4.2. Preparation of the horses

The experiment took place in a calm environment, in shade, without direct sunlight, in the same part of the day, at a similar ambient temperature. The data for the ambient conditions (ambient temperature and relative humidity) were retrieved through direct measurement via a meteorological station (brand: All Weather, Oregon Scientific®).

The horses were groomed to remove any dirt that could change the thermographic measurements, and afterwards, they spent 30 minutes in a thermally neutral box, without bedding, direct sunlight, or movement of air, in order to stabilise their skin temperature. During this period, the treated anatomical region was not touched.

The treatment area was determined by delimiting a rectangular area in the lateral part of the metacarpal region. As already noted, the animals were kept in a thermally neutral environment for 30 minutes: prior studies had shown that this period was necessary for the stabilisation of the skin temperature (Soroko et al., 2017; 2019).

4.3. Application of the bandages and ice protectors

Bandages were applied on the front right extremity, and ice protectors, on the front left extremity in 5 horses, and vice versa in the other 5 horses. This distribution was randomly assigned.

The Incrediwear bandages were soaked in a bucket of water at an average temperature of 20 °C for 20 minutes.



Figure 2 and 3: Incrediwear Equine® bandages soaked in a bucket of water and used to create the cryotherapeutic 'cold effect'; photographs by Marta Cobo.

They were subsequently applied to the corresponding extremity (left or right, depending on the horse), as shown in Figures 5, 6, and 7.



Figure 4 and 5: left: Incrediwear bandages. Right: Incrediwear bandages applied to a front extremity of one of the studied horses (source: Incrediwear bandage website; photograph by Marta Cobo).

The ice protectors had been kept in a freezer for at least 5 hours to ensure that they were frozen.



Figure 6 and 7: Conventional ice protectors. Photographs by Marta Cobo.

4.4. Thermographic camera and measurement procedure

For the thermal measurements, a FLIR® (Forward-Looking Infrared) thermographic camera, model i7, was used, with a thermal sensitivity (NETD) of $<0.1\text{ }^{\circ}\text{C}$ at $25\text{ }^{\circ}\text{C}$, FOL 7 mm lens, IR resolution 140×140 , and an emissivity set at 0.96. This thermographic camera model measures temperatures of up to $+250\text{ }^{\circ}\text{C}$, detecting temperature differences exceeding $0.10\text{ }^{\circ}\text{C}$. The precision of the device is $\pm 2\text{ }^{\circ}\text{C}$, or $\pm 2\%$, which means that it generates images with a sensitivity sufficient for the necessities of our study. The camera saved the images in JPEG format, for subsequent processing and temperature measurements.



Figure 8: FLIR i7 thermographic camera. Left photograph by Marta Cobo Moreno; right photograph retrieved from the website of an online shop.

The thermographic camera was situated at a distance of one metre from the animal, making sure that the position of the camera was perpendicular during the production of the images, given that the angle of the camera has a significant modifying effect on the measurement of temperature (Soroko et al., 2019; MacKechnie-Guire et al., 2021). For this reason, a tripod was used, making sure that the animal moved as little as possible.

The thermographic measurements were performed at the following times: before application (used as the baseline value without the bandages or ice protectors), immediately after removal (after applying the bandages or ice protectors and keeping them on for 20 minutes, within the first minute after their removal), and then at 5, 10, 15, and 30 minutes after removal. During these 30 minutes, the horses remained in the same thermally neutral box where the baseline measurement was performed.

Three thermal images were taken at every measurement time. The presented values were measured based on 3 consecutive images.

4.5. Statistical study

The data are presented as median and quartile values. The normal distribution of the data was assessed using the Shapiro-Wilk test (W). The W value for Tmax was significant, refuting the assumption that the data had a normal distribution.

The comparative study of the different measurement times for both the cryotherapy treatments analysed (before application, immediately after removal, and then at 5, 10, 15, and 30 minutes after removal) was performed using a Friedman test for repeated samples, and then a Wilcoxon test was used as post-hoc analysis.

The comparison between the two cryotherapy systems (bandages and ice protectors) for each of the measurement times was assessed using a Mann-Whitney test.

In all the cases, the level of significance found was at $p < 0.05$. The Statistica for Windows software was used (v. 13.0).

5. Results

The thermal images of the horses at different measurement times are shown in Annex 1.

Table 1 shows the median and percentile Tmax values at different measurement times, comparing the two cryotherapy procedures.

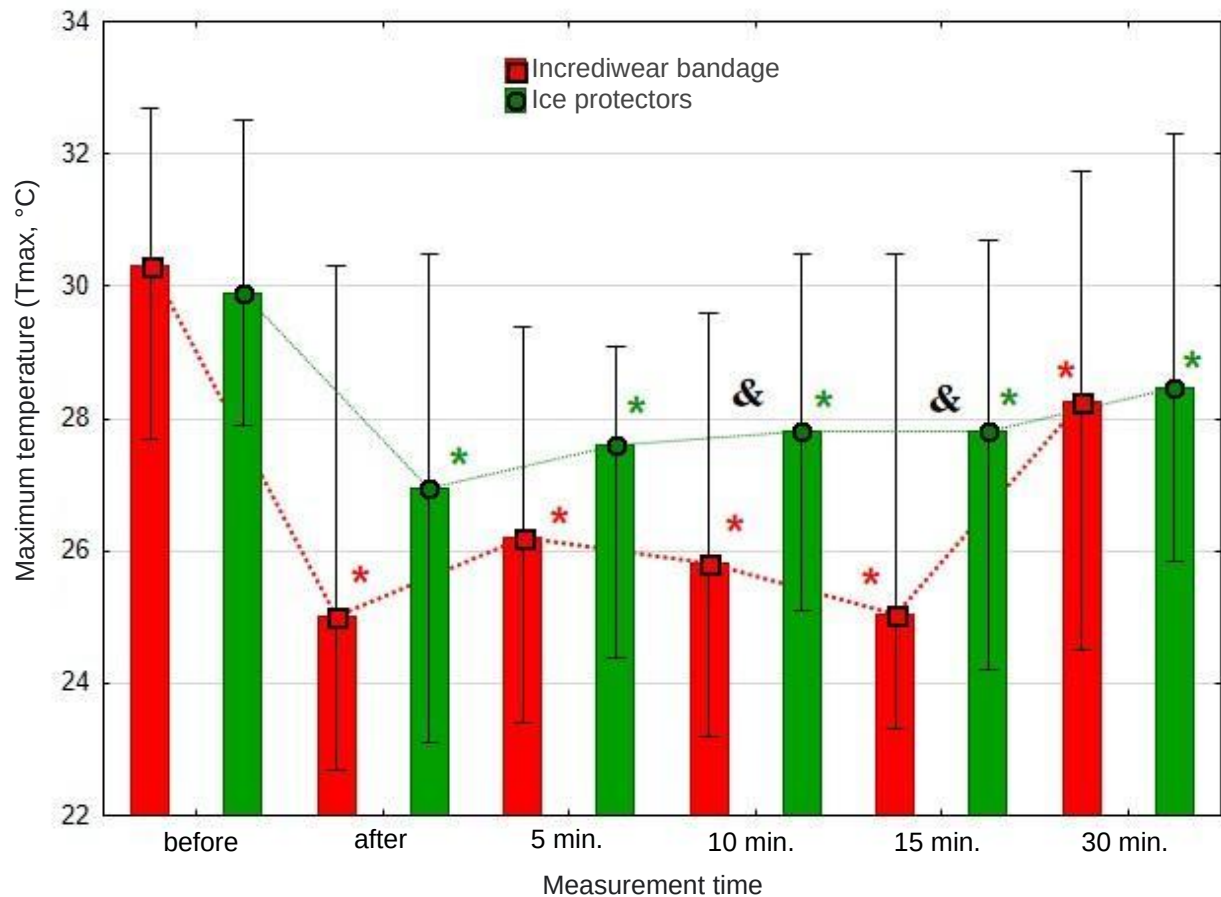


Table 1. Median and quartile value of the maximum temperature (°C) at different measurement times in 10 horses, using two different cryotherapy systems (Incrediwear bandages, ice protectors).

(red asterisks: significant differences relative to the Tmax measured before the application of the Incrediwear bandage;
green asterisks: significant differences relative to the Tmax measured before the application of the ice protectors;
ampersands: significant differences between the Incrediwear bandage and the ice protectors at the given measurement time).
Significant differences at $p < 0.05$

As can be seen in Table 1, the application of either of the cryotherapy systems resulted in a significant reduction of Tmax at all measurement times, as compared to the Tmax measured before the application of the cryotherapy systems. With the Incrediwear bandages a lower Tmax was measured at 15 minutes after removing them, while with the ice protectors, a lower Tmax was measured immediately after the removal of the protector.

In terms of the differences between the two systems, it was noted that when the Incrediwear bandages were used, the Tmax value was lower at 10 and 15 minutes after the removal of

the two systems.

In order to better understand the result, we calculated the percentage differences in Tmax, as compared to the baseline value, measured before the application of the two systems. Expressing these values as percentages revealed a sharper decrease in Tmax for the Incrediwear bandages, as opposed to the ice protectors, which is shown in Table 2, although no statistical analysis of these data was carried out.

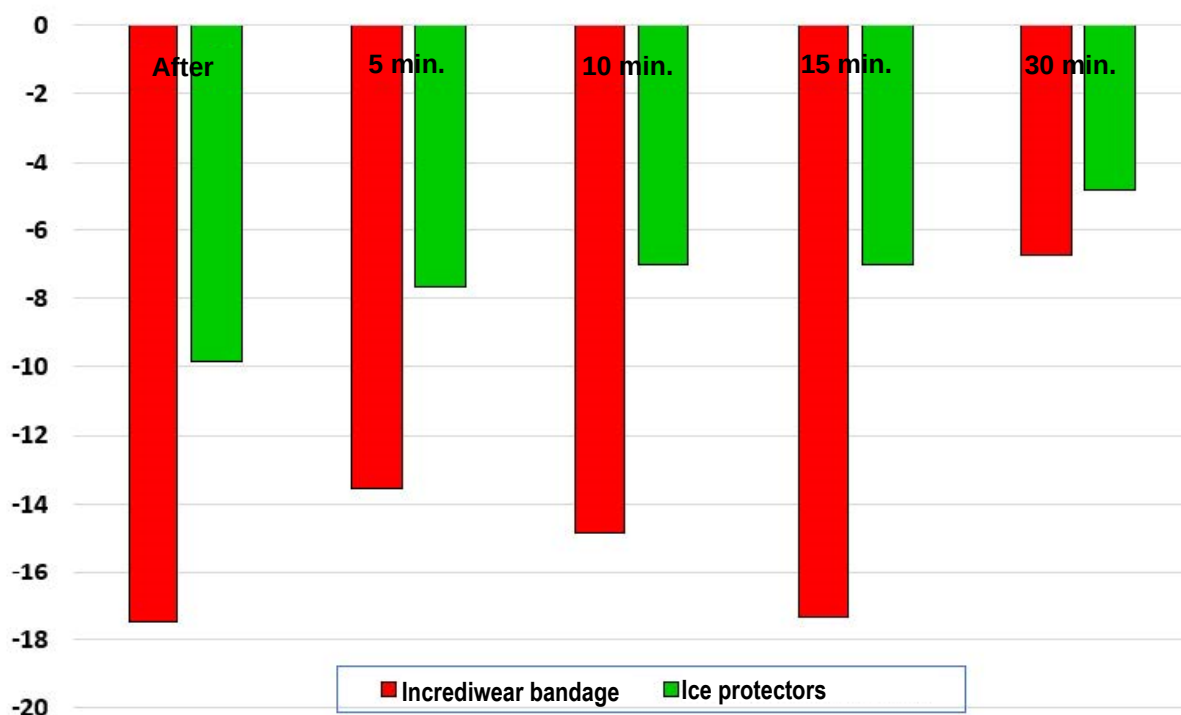


Table 2. Tmax percentage change relative to the baseline values (measured before the application of the two cryotherapy systems).

6. Discussion

Damage to the soft tissues of the metacarpal and metacarpophalangeal regions is very frequent in sports horses. In show jumping horses, injuries of the SDFT, deep digital flexor tendon, and fetlock suspensory ligament of the front limbs are frequent, with a higher incidence in elite horses, as compared to non-elite, being proportional to the height of the obstacles (Murray et al., 2002). Likewise, it has been reported that the frequency of training for show jumping and the height of the obstacles determine the occurrence of tendinopathies in the metacarpal, metacarpophalangeal, and pastern regions (Meershoek et al., 2001; Egenvall et al., 2013).

The excess weight stress that these structures undergo when the horse lands after a jump is significant, and the resulting trauma leads to progressive degeneration. For this reason, cryotherapy has been suggested as a useful treatment for reducing the risk of injuries, controlling post-exercise inflammation and limiting accumulating degenerative processes that lead to injuries (Dakin et al., 2014). Actually, it is believed that cryotherapy reduces pain, swelling, and inflammation through various mechanisms, which include vasoconstriction, reduction in cell metabolism demand, reduction in the inflammatory mediators, lower nerve conduction speed, and higher collagen rigidity. Furthermore, it is considered to provide a beneficial local analgesic effect. It is hoped that the temperatures achieved in SDFT through the use of cryotherapy could create a clinically beneficial effect on the inflamed tissues.

It is common for many riders and coaches to use ice protectors or conduct hydrotherapy sessions, applying water to the metacarpal region after the horse completes a period of intense exercise. The price of these protectors is approximately 100 euros per anatomical area. There are also ice protectors that apply vibration (costing approximately 250 to 300 per unit) or more advanced systems, with the combined application of cryotherapy and compression, for which devices cost some 3000 euros, plus a protector or application system for each anatomical area. In recent years, the company Incrediwear has begun marketing germanium and carbon bandages in a form similar to that intended for humans, with a price of 40 to 90 US dollars per pair. This manufacturer claims that due to the properties of germanium and carbon, after these bandages are soaked in water they effect a reduction in temperature. So far, to the best knowledge of the authors of this study, there have been no evaluations of the temperature changes created by these bandages in comparison to other methods more common in equine sports practice, such as the use of ice protectors.

The two hypotheses proposed in this study were supported. First, we hypothesised that the application of these bandages soaked in water at ambient temperature (20 °C) would reduce the temperature of the metacarpal region. In fact, we saw that the application of the bandages led to a reduction of 17.49%. Second, we hypothesised that the Incrediwear bandages would have a more substantial effect on temperature relative to the baseline value (before application), compared to the ice protectors. This second hypothesis was also supported, as we demonstrated significant differences between the two systems, which manifested at 10 and 15 minutes after their removal. Even though the reduction in temperature was more notable with Incrediwear bandages, significant differences from ice protectors were not found across all measurement times. The reason for this could be variation among the animals, with elevated standard deviations (these values are not shown), which are between 3 and 6: these values are within the difference between the median values obtained for the two systems.

The possible causes for this individual variation could be: (1) amount of fur and subcutaneous tissue in each animal, that is, the thickness of the skin; (2) ambient conditions on the different days of the experiment.

The amount of fur was apparently very similar in all the animals, which makes us believe that it did not affect the study. The studied horses belonged to different breeds, which could have had an effect, although we have not been able to estimate it. The authors are not aware of any studies that assessed the thickness of skin in different horse breeds. One must take into account the fact that all of these animals were sports horses, which suggests that the influence of this factor should be minimal.

Secondly, it could be presumed that there was a certain influence exerted by the ambient conditions on the measurement days. However, we do not believe that this variation in ambient conditions could have been the cause of the difference among the horses. The maximum and minimum values of the ambient temperature and relative humidity across all measurement days were 16.9 to 21.0 °C and 37 to 47%, respectively. The difference is small, especially for temperature, which in our opinion, does not explain the individual variation. Furthermore, we must point out again that the measurements of the two cryotherapy systems (Incrediwear bandages versus ice protectors) were performed on both of the front limbs at the same time: the study was thus designed to avoid the influence of ambient conditions.

This means that our study has confirmed that Incrediwear bandages cause a T_{max} reduction in the palmar metacarpal region that is more substantial than the reduction produced by ice protectors. Another very interesting result is that the lower T_{max} value persisted for a longer time with Incrediwear bandages, implying that the cold created by them was longer lasting.

We believe that bandages with germanium and carbon have practical advantages compared to ice. They are more adaptable to the anatomical shape of this region in every horse, they do not cause frost-induced skin damage, and we believe that the likelihood of chafing is lower in comparison with ice protectors. However, it is worth noting that these ideas are our unconfirmed estimations. One important advantage is the lack of need to have a freezer for cryotherapy. If one uses ice protectors, one obviously has to keep the ice pack frozen, which requires

a freezer. Only water is necessary for the bandages, something that is notable from a practical point of view in their use by riders and coaches.

Our study had a number of limitations. First, only T_{max} was measured. It would have been advisable to determine the average temperature as well as the number of pixels (indicating the area) with different temperatures. We only measured maximum temperature because of our use of the free software by the company Flir available on their website, which does not make it possible to repeatedly set the measurement area. For this reason, we chose to calculate T_{max}.

Secondly, this being a preliminary study, we only investigated horses at rest. We believe that if this study were also carried out with horses after an exercise session or a competition, the temperature difference at different measurement times could have been more evident. We plan to do this in the near future.

Thirdly, we measured the temperature on the surface, which does not necessarily represent the internal temperature of the flexor tendons. We have found a number of articles that determine the internal temperature of the tendon through the surgical implantation of thermal probes (Montgomery et al., 2013; Haussler et al., 2021). This would require the use of experimental horses, not privately owned ones, for which reason, this approach would not have been viable in the design of our experiment, even if we consider using it in the future.

Fourthly, one must ask the question of how effective the reduction in T_{max} achieved by the two systems could be. According to Haussler et al. (2021), one needs a reduction down to 10–15 °C to maximise the physiological effects of vasoconstriction, pain relief, and metabolic rate. We do not know if the temperature reduction achieved could be sufficient from a therapeutic or preventative point of view.

Finally, we must be aware of the fact that although hyperthermia has been considered a risk factor in the development of tendinopathies, it is not the only one, meaning that controlling the temperature should not obviate other important measures, such as monitoring the intensity of the exercise, controlling the level of fatigue, optimising post-exercise recovery, correct horseshoes and running tracks, etc.

7. Conclusions

This study was designed to assess the reduction in the skin temperature of the metacarpal

palmar region in resting horses after the application of two cryotherapy systems: bandages manufactured using germanium and carbon and soaked in water, and ice protectors.

The main conclusions we found are as follows:

FIRST CONCLUSION. The application of bandages impregnated with germanium and carbon and the application of ice protectors in the metacarpal region resulted in a significant reduction of the skin temperature, as measured thermographically, which remained below the baseline temperature for 30 minutes after the removal of the two cryotherapy systems.

SECOND CONCLUSION. The application of bandages impregnated with germanium and carbon effected a higher reduction of the maximum temperature of the metacarpal palmar region than the application of ice protectors, as measured at 10 and 15 minutes after the removal of the two systems.

THIRD CONCLUSION. The lowest maximum temperature values were achieved 15 minutes after the removal of the bandages impregnated with germanium and carbon and immediately after the removal of the ice protectors. This suggests that the bandages impregnated with germanium and carbon have a better capacity for maintaining a lower temperature for a longer period, in comparison to ice protectors.

8. List of references

9. Acknowledgements

I would like to thank my parents, Diego and Rosana, for making it possible for me to spend 4 years studying the world of horses, and then launch myself into it completely, never to leave, in addition to letting me develop as an athlete, reaching a level that we would never have dreamed of.

Thanks to my university classmates. They're my foundation. I am so lucky to know them. This has been the beginning of a long road for us together.

Thank you, Alberto. For your enthusiasm, for your support, and for being my motivation. I hope that I can always make you proud of me.

Thanks to my research advisors Ana and Paulo. Thank you for your time and dedication, it has been a pleasure, and I hope to be able to do another project as exciting as this one.

ANNEX 1. Thermal images of a horse that participated in the study.

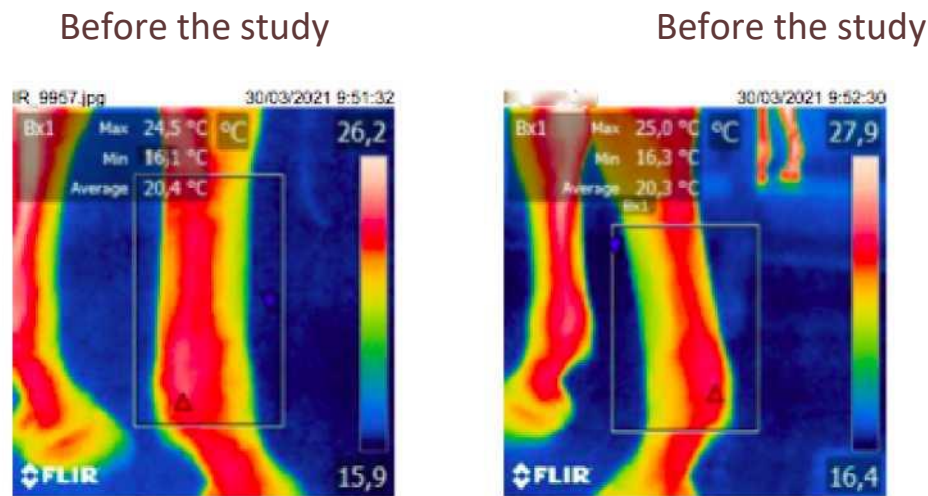


Figure 11: Thermal images of the front extremities of one of the studied horses, before the use of the bandages or ice protectors, taken to determine the baseline temperature. Photographs by Marta Cobo.

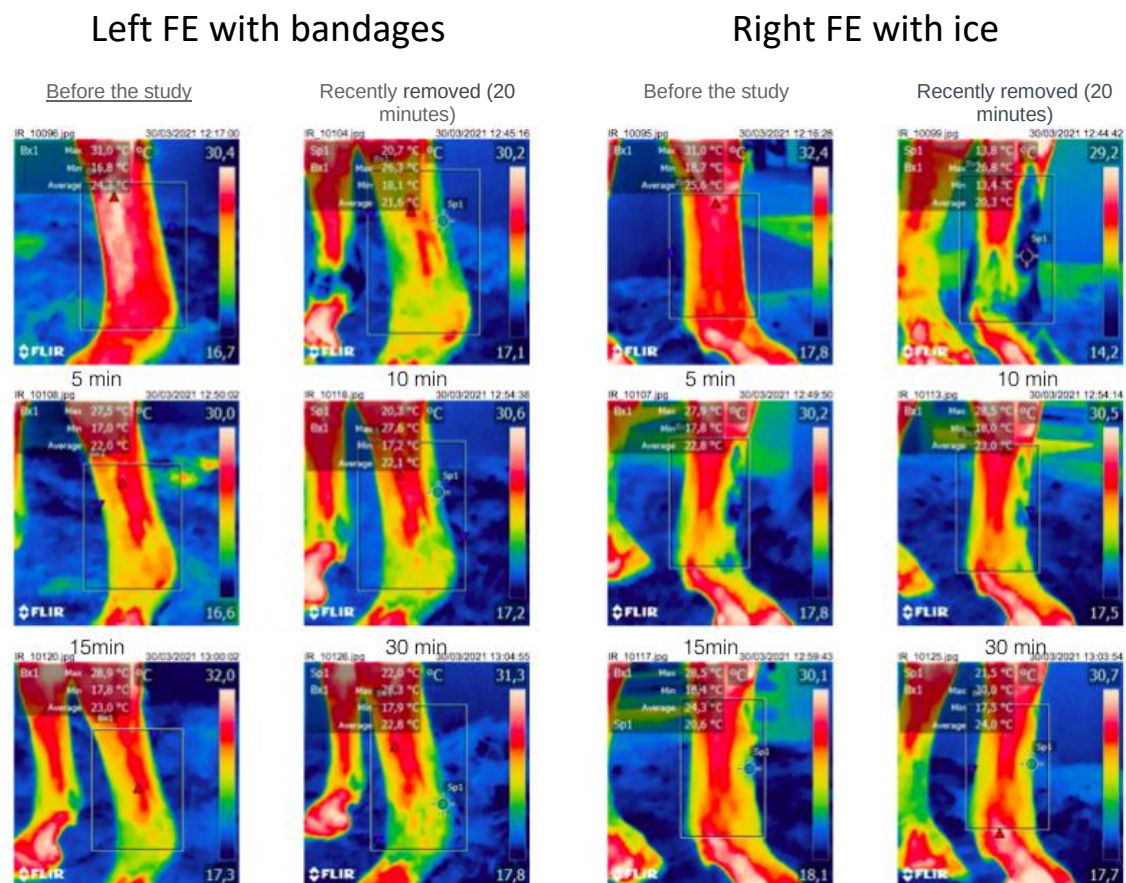
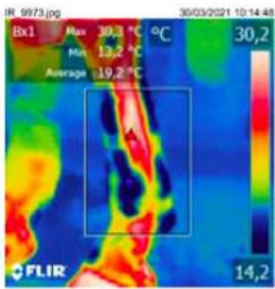


Figure 12: Thermal images of the front extremities of one of the studied horses. Surface temperature measured before the application of the bandages or ice protectors, immediately after their removal (after keeping them applied for 20 minutes), and at 5, 10, 15, and 30 minutes after their removal.

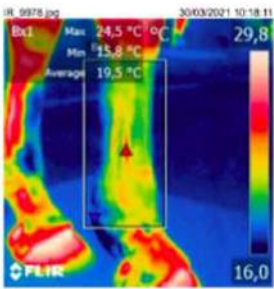
Recently removed, after 20 minutes

Recently removed, after 20 minutes

Left FE with ice



Right FE with
bandages



After 30 minutes

After 30 minutes

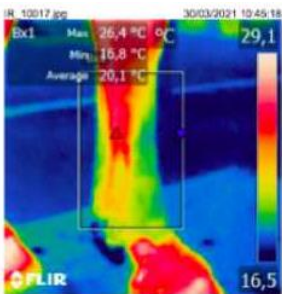
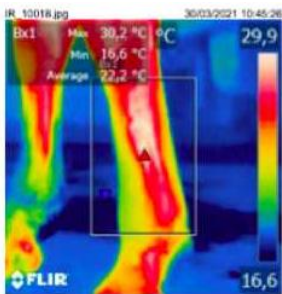
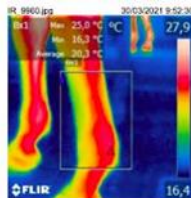
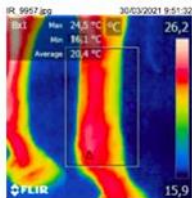


Figure 13: Thermal images showing the effects of the Incrediwear bandage as compared to the ice protectors. Photographs by Marta Cobo.

Right FE with ice and left FE with bandages

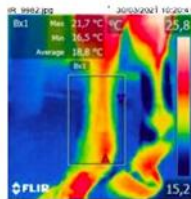
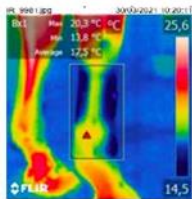
Before the study

Before the study



5 minutes after removal (20 minutes)

5 minutes after removal (20 minutes)



After 15 minutes

After 15 minutes

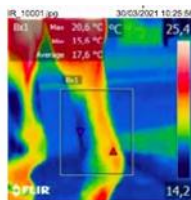
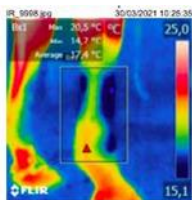


Figure 14: Thermal images comparing the Incrediwear bandages with the ice protectors. Photographs by Marta Cobo.